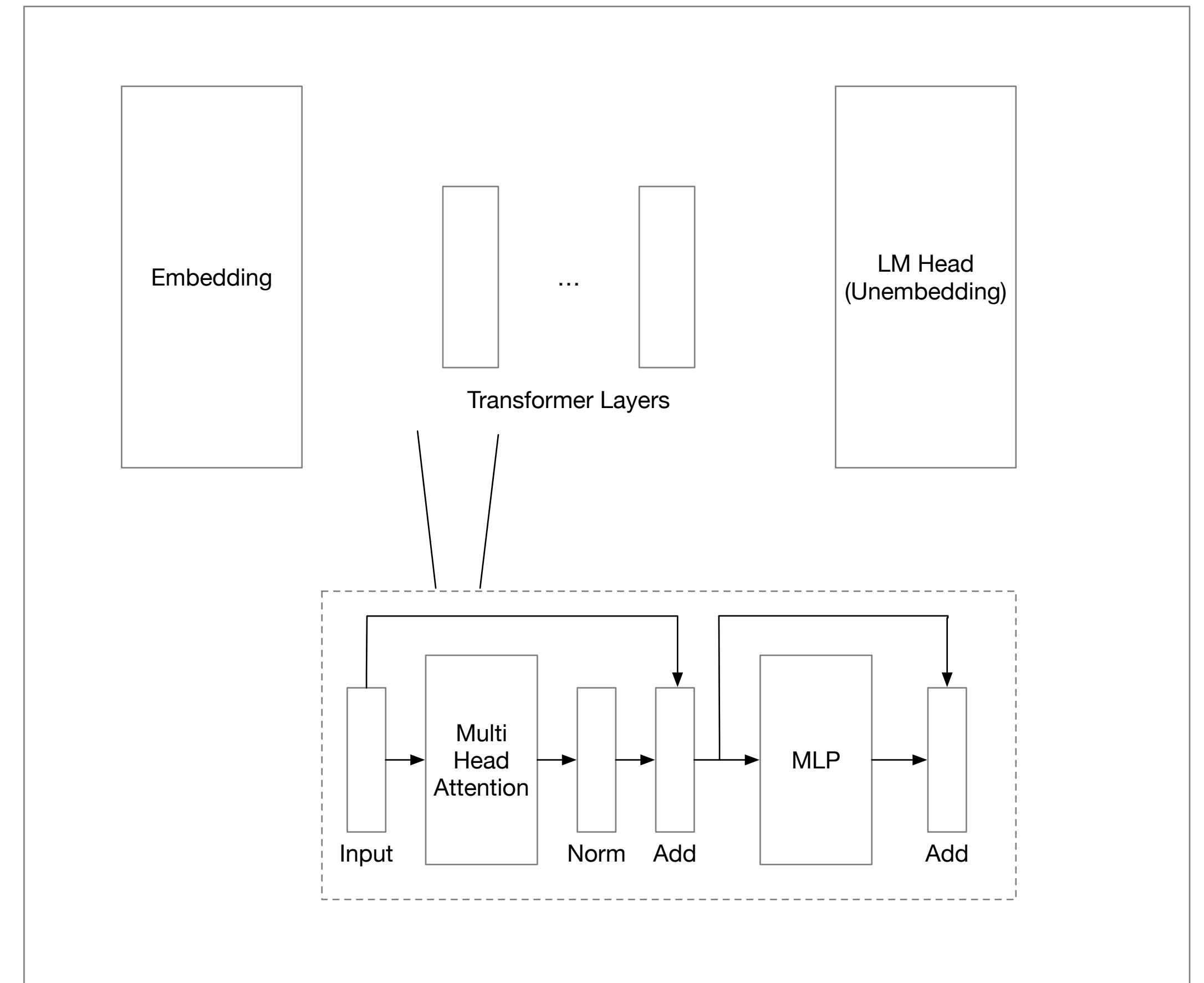


Fitting LLMs on iPhones

Quick Recap

Decoder only Transformers
aka GPT

- Input: a sequence of tokens
- Output: a probability distribution
- Embedding and Un-embedding layers
- Sandwich of transformer layers in between



Quick Recap

Self-attention

- Q, K and V linear projections
- Normalized by $\sqrt{d_k}$
- Each layer has multiple heads
- Outputs of heads are concatenated, followed by another matmult

$$ATTENTION(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

	Q ₁	Q ₂	Q ₃	Q ₄
K ₁	dotprod	dotprod	dotprod	dotprod
K ₂	∞	dotprod	dotprod	dotprod
K ₃	∞	∞	dotprod	dotprod
K ₄	∞	∞	∞	dotprod

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_n)W^o$$

$$\text{Where } head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$$

Quantization

- Embeddings, Transformer layers and LMHead are all just a bunch of matrices
- Each weight is a fp16/bf16 param (2 bytes).
- Naive quantization doesn't work very well.
- Group quantization does.

$$X_{int} = \left\lceil \frac{X_{float} - X}{S} \right\rceil$$

$$S = \frac{\max(X_{float}) - \min(X_{float})}{2^{N-1} - 1}$$

In symmetric quantization

$$Z = 0$$

In asymmetric quantization

$$Z = \min(X_{float})$$

Quantization

Group quantization

- Types
 - Data free
 - RTN (Round to nearest)
 - HQQ
 - Data based
 - GPTQ
 - AWQ
 - OmniQuant

Z	S	X_int	X_int	X_int	X_int
Z	S	X_int	X_int	X_int	X_int
Z	S	X_int	X_int	X_int	X_int
Z	S	X_int	X_int	X_int	X_int
Z	S	X_int	X_int	X_int	X_int

$GS = 4$

Quantization

- Quantization can be frame as an optimization problem.
- Data free quantization optimizes for reconstruction loss of quantization matrices.
 - $\|W - Dequant(Quant(W))\|_2$
- Data based quantization optimizes to minimize the difference between the representations resulting from unquantized matmult and quantized matmult.
 - $\|Wx - Dequant(Quant(W))x\|_2$

Final trick

- Reducing context length

Fin